

ORIGINAL ARTICLE

WILEY

Henagliflozin monotherapy in patients with type 2 diabetes inadequately controlled on diet and exercise: A randomized, double-blind, placebo-controlled, phase 3 trial

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Abstract

Aim: To evaluate henagliflozin, a novel sodium-glucose co-transporter-2 inhibitor, as monotherapy in patients with type 2 diabetes and inadequate glycaemic control with diet and exercise.

Materials and Methods: This multicentre trial included a 24-week, randomized, double-blind, placebo-controlled period, followed by a 28-week extension period. Four hundred and sixty-eight patients with an HbA1c of 7.0%–10.5% were randomly assigned (1:1:1) to receive once-daily placebo, or 5 or 10 mg henagliflozin. After 24 weeks, patients on placebo were switched to 5 or 10 mg henagliflozin, and patients on henagliflozin maintained the initial therapy. The primary endpoint was the change in HbA1c from baseline after 24 weeks.

Results: At Week 24, the placebo-adjusted least squares (LS) mean changes from baseline in HbA1c were -0.91% (95% CI: -1.11% to -0.72% ; $P < .001$) and -0.94% (-1.13% to -0.75% ; $P < .001$) with henagliflozin 5 and 10 mg, respectively; the placebo-adjusted LS mean changes were -1.3 (-1.8 to -0.9) and -1.5 (-2.0 to -1.1) kg in body weight, and -5.1 (-7.2 to -3.0) and -4.4 (-6.5 to -2.3) mmHg in systolic blood pressure (all $P < .05$). The trends of these improvements were sustained for an additional 28 weeks. Adverse events occurred in 81.0%, 78.9% and 78.9% of patients in the placebo, henagliflozin 5 and 10 mg groups, respectively. No diabetic ketoacidosis or major episodes of hypoglycaemia occurred.

Conclusions: Henagliflozin 5 mg and 10 mg as monotherapy provided effective glycaemic control, reduced body weight and blood pressure, and was generally well tolerated.

KEYWORDS

henagliflozin, sodium-glucose co-transporter-2 inhibitor, type 2 diabetes

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1 | INTRODUCTION

Diabetes is a global public health burden. According to the International Diabetes Federation report, in 2019, 463 million adults were estimated to have diabetes worldwide, with China accounting for approximately 25% of this total with 116 million adults (and a prevalence of 12.8%).^{1,2} Conventionally, the first-line treatment for type 2 diabetes (T2D), the predominant type of diabetes, is metformin in addition to lifestyle modification.³ Nevertheless, metformin is intolerable to some patients because of gastrointestinal events.⁴ Furthermore, because of the progressive nature of T2D, treatment intensification with the introduction of antidiabetic agents targeting complementary mechanisms is often required to achieve glycaemic targets.³ There remains a need for novel antidiabetic drugs, particularly ones that act independently of insulin.

Sodium-glucose co-transporter-2 (SGLT2) inhibitors are a comparatively new class of antidiabetic agents that primarily act via the kidney.⁵ SGLT2 inhibitors improve hyperglycaemia by blocking glucose reabsorption and thereby increase urinary glucose excretion. Because of this specific mode of action, SGLT2 inhibitors are associated with additional cardiometabolic benefits, including weight loss from increased glucosuria/loss of calories and a decrease in blood pressure from osmotic diuresis.⁵ Among the notable SGLT2 inhibitors, dapagliflozin, canagliflozin and empagliflozin have been approved for the treatment of adult patients with T2D. With recent findings from cardiovascular (CV) outcome trials showing the CV and renal protective effects of SGLT2 inhibitors,^{6–10} the 2020 American Diabetes Association guideline recommends the prioritized use of SGLT2 inhibitors in T2D patients with heart failure or chronic kidney disease, whereas the 2019 European Society of Cardiology/European Association for the Study of Diabetes guideline recommends SGLT2 inhibitors as first-line treatment for patients with CV disease or high CV risk before metformin.^{3,11}

Henagliflozin (SHR3824) is a novel, highly selective, oral inhibitor of SGLT2.^{12,13} Our previous phase 2 trial (NCT02366351) showed that 12-week treatment of henagliflozin monotherapy provided efficacious glycaemic control, reduced body weight and blood pressure, and was generally well tolerated in Chinese patients with T2D and

inadequate glycaemic control with diet and exercise (data submitted for publication). Herein, we conducted a phase 3 trial to further evaluate the efficacy and safety of henagliflozin monotherapy in this population, with results from the 24-week placebo-controlled period and additional 28-week extension period reported.

2 | MATERIALS AND METHODS

2.1 | Study design

This multicentre phase 3 study conducted at 36 centres in China consisted of a 4-week placebo run-in period, and a 24-week, double-blind, randomized, placebo-controlled period, followed by a 28-week extension period (Figure S1 in Appendix S1). The study was conducted in accordance with the Declaration of Helsinki and the International Conference on Harmonization/Good Clinical Practice Guidelines as well as local regulatory requirements. The study protocol was approved by the ethic committees of all the participating centres and all subjects provided written informed consent before enrolment. The study is registered with ClinicalTrials.gov (NCT03159052).

2.2 | Patients

Eligible patients were adults aged 18–75 years with a diagnosis of T2D according to the 1999 World Health Organization criteria; and who had body mass index (BMI) 19–35 kg/m²; had received diet and exercise treatment for 8 weeks or longer prior to screening and had not received antidiabetic medication within 8 weeks; had HbA1c of 7.5%–11.0% at screening (local test) and 7.0%–10.5% at randomization (central test); and had fasting plasma glucose (FPG) of 15 mmol/L or less at both screening (local test) and randomization (central test). Key exclusion criteria were type 1 diabetes or secondary diabetes; acute metabolic complications (e.g. ketoacidosis and lactic acidosis) within 6 months; history of repeated urinary tract and/or genital infections; history of a CV event within 6 months; uncontrolled

blood pressure (systolic blood pressure [SBP] >160 mmHg and/or diastolic blood pressure [DBP] >100 mmHg) despite drug treatment; or an estimated glomerular filtration rate of less than 60 mL/min/1.73m².

2.3 | Randomization and masking

Eligible patients at screening entered a 4-week, once-daily placebo run-in period. After the run-in, patients with sufficient treatment compliance who met the glycaemic criteria for inclusion were randomly assigned (1:1:1) to receive once-daily placebo, or henagliflozin 5 or

10 mg for 24 weeks. Patients in the placebo group were switched, in a 1:1 ratio of randomization generated during allocation at the beginning of the 24-week treatment period, to receive henagliflozin 5 or 10 mg during the 28-week extension period. Patient allocation was based on a computer-generated randomization code using an interactive web response system and was stratified by baseline HbA1c (<8.5% and >8.5%). Patients, investigators, evaluators and data analysts were blinded to the treatment assignment from the time of randomization to the locking of the database.

During the 24-week placebo-controlled treatment period, henagliflozin 5 or 10 mg and placebo were administered orally once daily in the morning. Patients who completed the 24-week period

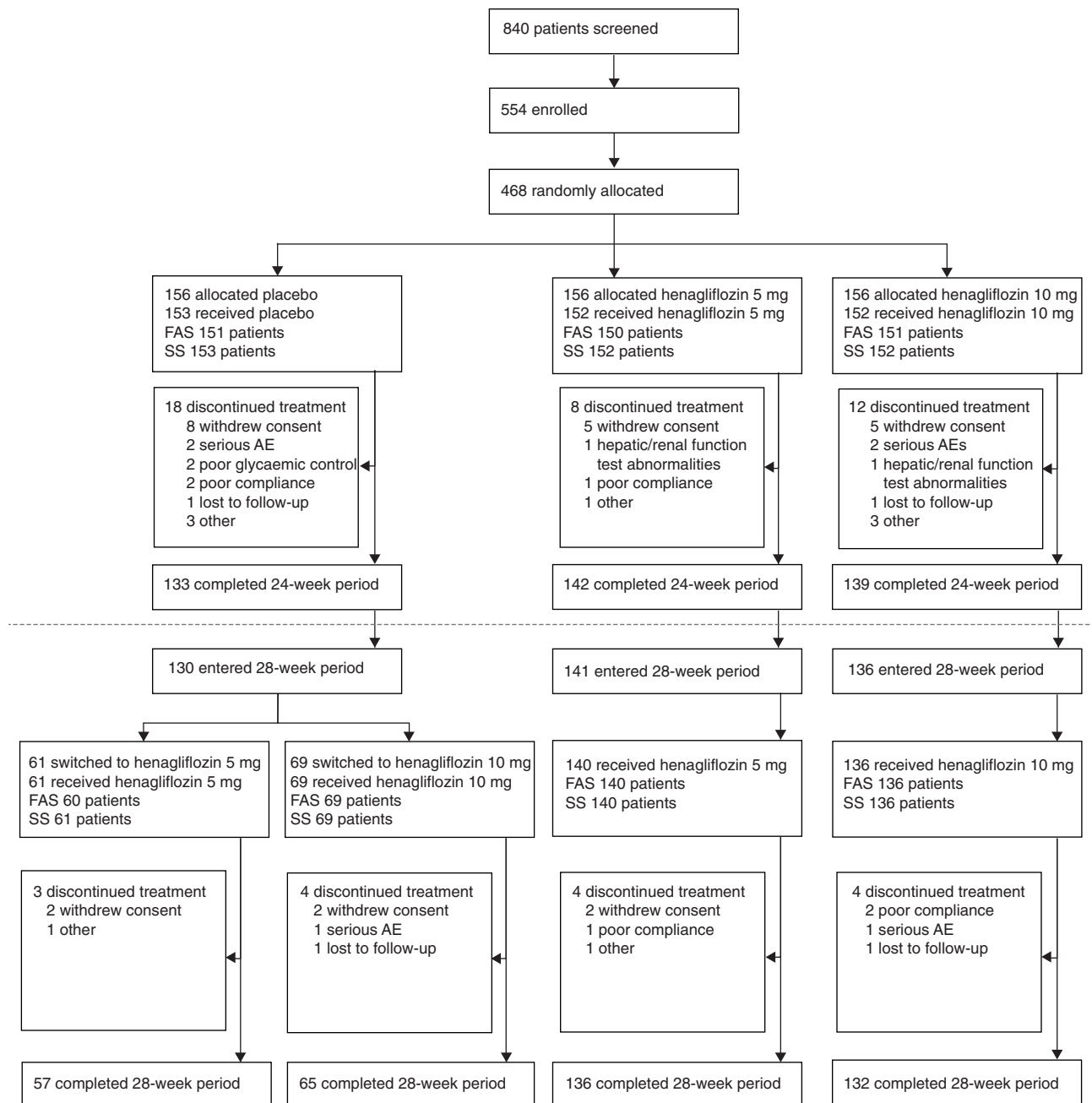


FIGURE 1 Trial profile. AE, adverse event; FAS, full analysis set; SS, safety set

were eligible to enter the 28-week extension period. During the extension period, the treatment for patients initially on henagliflozin remained unchanged whereas the treatment for patients initially on placebo was switched to henagliflozin 5 or 10 mg once daily according to the randomization scheme. Study visits were planned every 4 weeks during the 24-week placebo-controlled period and every 7 weeks during the 28-week extension period. Rescue medication was started if FPG was more than 15 mmol/L between Week 1 and Week 6, more than 13.3 mmol/L between Week 7 and Week 12, or more than 11.1 mmol/L between Week 13 and Week 52. These progressively more stringent glycaemic rescue criteria were used to maximize the safety of the participants.

2.4 | Endpoints and assessment

The primary efficacy endpoint was change from baseline in HbA1c at Week 24. Secondary efficacy endpoints were change from baseline in HbA1c at Week 52, the proportions of patients achieving an HbA1c of less than 6.5% and less than 7.0% at Week 24, and the changes from baseline in FPG, 2-hour postprandial glucose (2-h PPG) level, PPG area under the concentration-time curve (AUC_{0-2h}), body weight and blood pressure at Week 24 and Week 52. A meal tolerance test was used to assess 2-h PPG and PPG AUC_{0-2h} . Baseline and post-treatment HbA1c and glucose (fasting or postprandial) were all centrally measured.

Safety was monitored from the time of informed consent to the last patient visit and assessed with adverse event (AE) reports, laboratory tests, vital signs, physical examinations and electrocardiograms. After that, an additional post-treatment safety follow-up at 4 weeks was conducted by telephone. A major episode of hypoglycaemia was defined as symptomatic episodes requiring third party assistance and/or with a capillary or plasma glucose less than 3.0 mmol/L.

2.5 | Statistical analyses

Efficacy was assessed in the full analysis set (FAS), comprising all randomized patients who received at least one dose of study drug and had both baseline and at least one post-treatment measurement of the efficacy endpoint during the 24-week placebo-controlled period or 28-week extension period. The safety set (SS) included all enrolled patients who received at least one dose of study drug and had at least one post-treatment safety assessment in the placebo-controlled period or extension period. For patients who received rescue medication, data collected after the initiation of rescue therapy were not used for efficacy analysis but were included for safety analysis.

Sample size calculation was based on comparison of henagliflozin versus placebo in the change from baseline in HbA1c at Week 24. We calculated that 450 patients (150 in each of the placebo, henagliflozin 5 mg and henagliflozin 10 mg groups) would provide 97% power to detect a reduction in HbA1c of 0.55% in the henagliflozin 10 mg group versus placebo and 94% power to detect a reduction of 0.50% in the henagliflozin 5 mg group versus placebo based on a t-test at a two-sided type I error of 0.05, on the assumption of a common standard deviation (SD) of 1.1% and a dropout rate of 20%. An overall power of 91% was ensured for this study. For primary endpoint analysis, a hierarchical testing procedure was employed to strictly control the two-sided type I error at 0.05. The superiority of henagliflozin versus placebo was tested using a fixed-sequence step-down approach from high to low dose. If the superiority of henagliflozin 10 mg was established, a further test for henagliflozin 5 mg would be conducted.

The primary endpoint and continuous secondary endpoints at Week 24 were analysed using an ANCOVA model with treatment as fixed effect and the respective baseline value as a covariate. Missing data were imputed using the last observation carried forward method for HbA1c, FPG, body weight and blood pressure. Two-hour PPG and AUC_{0-2h} were analysed using the observed case approach. Least squares (LS) means

	Placebo (n = 151)	Henagliflozin 5 mg (n = 150)	Henagliflozin 10 mg (n = 151)
Age (years)	52.4 ± 10.2	53.3 ± 9.6	52.2 ± 9.4
Men, n (%)	100 (66.2)	88 (58.7)	115 (76.2)
Body weight (kg)	71.6 ± 10.6	70.1 ± 12.5	70.4 ± 11.8
BMI (kg/m ²)	26.0 ± 2.9	25.4 ± 3.1	25.5 ± 3.1
Diabetes duration (months)	26.5 (7.6, 52.2)	22.1 (4.7, 50.5)	23.5 (6.5, 49.8)
HbA1c (%)	8.63 ± 0.89	8.59 ± 0.89	8.59 ± 0.91
FPG (mmol/L)	9.83 ± 2.01	9.90 ± 2.05	9.88 ± 2.17
2-h PPG (mmol/L)	16.49 ± 3.15	16.86 ± 3.36	16.61 ± 3.34
LDL-C (mmol/L)	3.26 ± 0.83	3.19 ± 0.80	3.14 ± 0.88
eGFR (mL/min/1.73m ²)	124 ± 30	120 ± 32	122 ± 31

Abbreviations: BMI, body mass index; eGFR, estimated glomerular filtration rate; FPG, fasting plasma glucose; LDL-C, low-density lipoprotein cholesterol; PPG, postprandial glucose.

Note: Continuous data are mean ± standard deviation or median (interquartile range).

TABLE 1 Demographics and baseline characteristics in the full analysis set

and 95% CIs were calculated for the change from baseline within each treatment group and for the difference in change from baseline between groups. Prespecified subgroup analysis by baseline HbA1c (>8.5% vs. ≤8.5%), age (≥65 vs. <65 years), gender (male vs. female), BMI (≥24 vs. <24 kg/m²) and disease duration (< median vs. ≥ median) for the primary

endpoint was also performed. Category efficacy endpoints at Week 24 were compared using the Cochran-Mantel-Haenszel test stratified by baseline HbA1c (>8.5% or ≤8.5%), and non-response imputation was implemented for patients with missing data. The 95% CIs for category variables were estimated by the Clopper-Pearson method (absolute

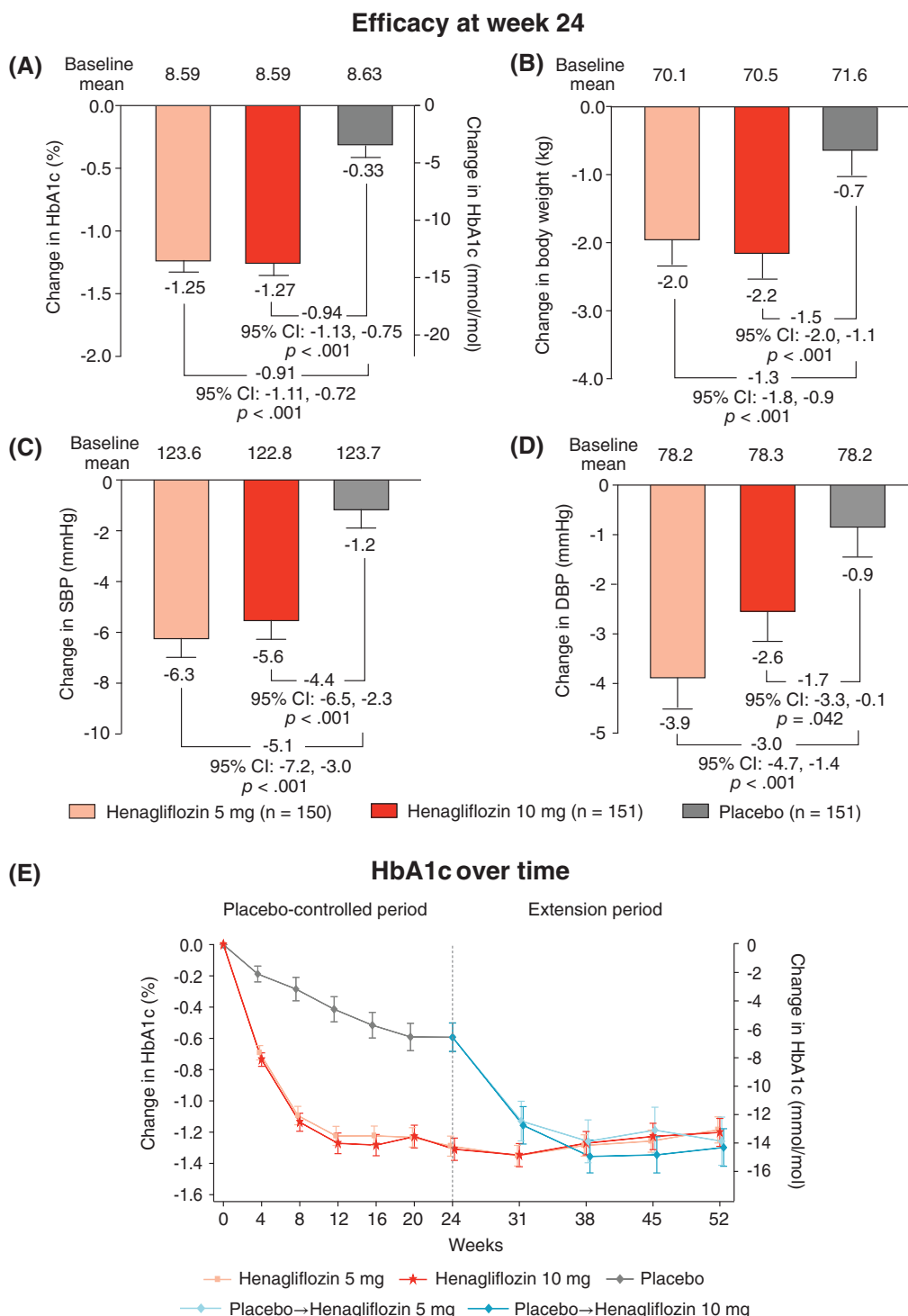


FIGURE 2 Change from baseline in metabolic variables. A, Change in HbA1c at Week 24. B, Change in body weight at Week 24. C, Change in SBP at Week 24. D, Change in DBP at Week 24. E, Change in HbA1c over 52 weeks. Data are least squares mean (LOCF) ± SE (A-D) or observed mean ± SE (E). *p*-values are adjusted for baseline values using ANCOVA. DBP, diastolic blood pressure; FPG, fasting plasma glucose; LOCF, last observation carried forward; SBP, systolic blood pressure; SE, standard error

proportion) or normal approximation (difference in proportion). For the extension period, only descriptive analyses were performed for all efficacy endpoints using the observed case approach. All statistical analyses were conducted with SAS 9.4 (SAS Institute Inc., Cary, NC, USA).

3 | RESULTS

From 16 June 2017 to 7 December 2018, 468 patients were enrolled and randomized, of whom 414 (88.5%; 85.3%–91.0% across groups) completed the 24-week placebo-controlled period. Of the 407 patients who entered the 28-week extension period, 390 (95.8%; 93.4%–97.1% across groups) completed it (Figure 1). In the 24-week period, 452 patients were included in the FAS and 457 in the SS; in the 28-week extension period, 405 were included in the FAS and 406 in the SS. The most common reason for treatment

discontinuation was patient withdrawal in both the 24-week placebo-controlled period (18/452; 4.0%) and 28-week extension period (6/405; 1.5%). During the study, 32 patients (7.1%; 30 [19.9%] of 151 in the placebo group) in the 24-week period and three patients (0.7%) in the 28-week period received rescue medication (mainly metformin, 29/32 and 2/3).

Baseline demographic and disease characteristics were balanced across all groups (Table 1). Overall, the mean age of the patients was 52.6 years, and 67.0% were male. The mean baseline HbA1c was 8.6% and the median duration of T2D was 23.2 months.

3.1 | Efficacy

At Week 24, significant improvement in the change from baseline in HbA1c was observed for both henagliflozin 5 mg (LS mean difference:

TABLE 2 Secondary endpoints at Week 24 in the full analysis set

	Henagliflozin 5 mg (n = 150)	Henagliflozin 10 mg (n = 151)
Change from baseline in FPG (mmol/L)		
LS mean difference vs. placebo (95% CI) ^a	−1.94 (−2.29, −1.59)	−2.16 (−2.51, −1.81)
P-value	<.001	<.001
Change from baseline in 2-h PPG (mmol/L)		
LS mean difference vs. placebo (95% CI)	−2.96 (−3.55, −2.38)	−2.69 (−3.27, −2.10)
P-value	<.001	<.001
Change from baseline in PPG AUC _{0-2h} (mmol*h/L)		
LS mean difference vs. placebo (95% CI)	−4.60 (−5.47, −3.73)	−4.50 (−5.37, −3.63)
P-value	<.001	<.001
Patients with HbA1c <6.5%		
Difference vs. placebo (95% CI) ^b	4.7 (−1.5, 10.9)	4.0 (−2.1, 10.1)
P-value	.109	.179
Patients with HbA1c <7.0%		
Difference vs. placebo (95% CI) ^b	20.8 (11.5, 30.0)	22.5 (13.2, 31.8)
P-value	<.001	<.001
Change from baseline in body weight (kg)		
LS mean difference vs. placebo (95% CI) ^a	−1.3 (−1.8, −0.9)	−1.5 (−2.0, −1.1)
P-value	<.001	<.001
Change from baseline in SBP (mmHg)		
LS mean difference vs. placebo (95% CI) ^a	−5.1 (−7.2, −3.0)	−4.4 (−6.5, −2.3)
P-value	<.001	<.001
Change from baseline in DBP (mmHg)		
LS mean difference vs. placebo (95% CI) ^a	−3.0 (−4.7, −1.4)	−1.7 (−3.3, −0.1)
P-value	<.001	.042

Abbreviations: AUC, area under curve; DBP, diastolic blood pressure; FPG, fasting plasma glucose; LS, least squares; SBP, systolic blood pressure; SD, standard deviation.

Note: Treatment effects were compared using Cochran–Mantel–Haenszel test for category variables and ANCOVA (baseline value as covariate) for continuous variables.

^aThe last observation carried forward method was used for handling missing values at Week 24.

^bNon-response imputation was used for handling missing values at Week 24.

TABLE 3 Adverse events during the 24-week treatment period in the safety set

	Placebo (n = 153)	Henagliflozin 5 mg (n = 152)	Henagliflozin 10 mg (n = 152)
AE	124 (81.0)	120 (78.9)	120 (78.9)
Treatment-related AE	38 (24.8)	53 (34.9)	59 (38.8)
Serious AE	10 (6.5)	6 (3.9)	6 (3.9)
Treatment-related serious AE	0	0	1 (0.7)
AE leading to study discontinuation	3 (2.0)	2 (1.3)	6 (3.9)
AEs occurring in $\geq 2\%$ patients in any group and more frequently with henagliflozin ^a			
Blood PTH increased	9 (5.9)	16 (10.5)	14 (9.2)
Urine ACR increased ^b	6 (3.9)	11 (7.2)	6 (3.9)
Diabetic ketosis	3 (2.0)	4 (2.6)	7 (4.6)
Blood ketone body increased	1 (0.7)	3 (2.0)	9 (5.9)
Toothache	2 (1.3)	5 (3.3)	6 (3.9)
eGFR decreased ^b	2 (1.3)	8 (5.3)	1 (0.7)
Albumin urine present ^b	1 (0.7)	5 (3.3)	4 (2.6)
Ketosis	0	2 (1.3)	8 (5.3)
Blood cholesterol increased	0	1 (0.7)	5 (3.3)

Abbreviations: ACR, albumin/creatinine ratio; AE, adverse event; eGFR, estimated glomerular filtration rate; PTH, parathyroid hormone.

Note: Data are n (%).

^aAll AEs occurring more frequently in the henagliflozin groups (either dose group) with a difference of $\geq 2\%$ vs. placebo are listed.

^bPatients partially overlapped in these categories.

-0.91% [95% CI: -1.11% to -0.72%], $P < .001$) and henagliflozin 10 mg (LS mean difference: **-0.94%** [95% CI: -1.13% to -0.75%], $P < .001$) compared with placebo (Figure 2A). Further subgroup analysis showed that the LS mean reductions in HbA1c at Week 24 were consistently more pronounced in patients receiving henagliflozin (5 or 10 mg) than in patients receiving placebo across all prespecified subgroups. For patients with a baseline HbA1c of 8.5% or less, the placebo-adjusted LS mean changes in HbA1c at Week 24 for the henagliflozin 5 and 10 mg groups were -0.66% (95% CI: -0.89% to -0.44%; $P < .001$) and -0.47% (95% CI: -0.70% to -0.25%; $P < .001$), respectively. **For patients with a baseline HbA1c of more than 8.5%, the placebo-adjusted LS mean changes in HbA1c at Week 24 for the henagliflozin 5 and 10 mg groups were -1.18% (95% CI: -1.48% to -0.88%; $P < .001$) and -1.43% (95% CI: -1.73% to -1.12%; $P < .001$), respectively.** Other secondary glycaemic variables at Week 24, including the proportion of patients achieving an HbA1c of less than 7.0%, reductions from baseline in FPG, 2-h PPG and PPG AUC_{0-2h}, were all more favourable with henagliflozin versus placebo (all $P < .001$), as shown in Table 2.

At Week 24, a reduction in body weight from baseline was observed for all three treatment groups, with a greater reduction noted in the henagliflozin 5 mg (LS mean difference: -1.3 [95% CI: -1.8 to -0.9] kg, $P < .001$) and 10 mg groups (LS mean difference: -1.5 [95% CI: -2.0 to -1.1] kg, $P < .001$) compared with placebo (Figure 2B). Similarly, a greater reduction in blood pressure was observed for both doses of henagliflozin over placebo at Week 24 (Figure 2C,D). The placebo-adjusted LS mean changes for henagliflozin 5 and 10 mg in SBP were -5.1 (95% CI: -7.2 to -3.0)

and -4.4 (95% CI: -6.5 to -2.3) mmHg, respectively; and the placebo-adjusted LS mean changes in DBP were -3.0 (95% CI: -4.7 to -1.4) and -1.7 (95% CI: -3.3 to -0.1) mmHg, respectively (all $P < .05$).

The results for efficacy endpoints **at Week 52** are presented in Table S1 in Appendix S1. In patients who remained on henagliflozin during the extension period, the mean ($\pm$ SD) changes **at Week 52 from baseline in HbA1c for henagliflozin 5 and 10 mg were -1.19% $\pm$ 0.96% and -1.20% $\pm$ 1.02%, respectively;** in patients who switched to henagliflozin from placebo, the mean changes were -1.26% $\pm$ 1.04% and -1.30% $\pm$ 0.87%, respectively (Figure 2E). Similarly, the reductions in other glycaemic variables (FPG, 2 h-PPG and PPG AUC_{0-2h}), body weight and blood pressure were maintained at Week 52 from Week 24 in patients who continued henagliflozin treatment during the extension period. In patients who switched to henagliflozin during the extension period, these variables were all reduced compared with the respective baseline values (Figure S2).

3.2 | Safety

Safety data up to Week 24 are presented in Table 3. The overall incidence of AEs was similar across all treatment groups, ranging from 78.9% to 81.0%. Average treatment compliance was high for all three treatment groups at 97.7%-98.1%. The rate of treatment discontinuation attributed to AEs was generally low, with 1.3%, 3.9% and 2.0% observed for henagliflozin 5 mg, henagliflozin 10 mg and placebo,

respectively. Serious AEs were reported for six patients (3.9%) in each of the henagliflozin groups, and for 10 patients (6.5%) in the placebo group. Among these, none were definitely or possibly related to treatment, and the causality of one serious AE (i.e. acute hepatic injury in the henagliflozin 10 mg group) was judged as unassessable by the investigator. No deaths were reported in the study.

AEs occurring more frequently in the henagliflozin groups versus placebo mainly included ketosis-related events and transient laboratory test abnormalities (Table 3). The frequency of urinary tract infections (4.6%, 3.3% and 3.9%, respectively) and hypoglycaemia (1.3%, 2.6% and 2.6%, respectively) was similar in the henagliflozin 5 mg, henagliflozin 10 mg and placebo groups. There were no cases of diabetic ketoacidosis (DKA) or major episodes of hypoglycaemia and only one case of hypotension (in the henagliflozin 10 mg group). The complete list of AEs occurring in 2% or more of patients in any group up to Week 24 is provided in Table S2.

The safety data regarding the extension period are provided in Table S3. In general, the main type, severity and frequency of AEs in patients on henagliflozin (remained on or switched to) during the 28-week extension period were consistent with those in patients on henagliflozin during the initial 24-week treatment period.

4 | DISCUSSION

Henagliflozin was the first SGLT2 inhibitor independently developed in China and this was the first phase 3 trial of an SGLT2 inhibitor in an all-Chinese T2D population. In patients with T2D and inadequate glycaemic control with diet and exercise, once-daily henagliflozin 5 or 10 mg as monotherapy resulted in significant and clinically meaningful reductions in HbA1c, body weight and blood pressure after the 24-week treatment period; these benefits were maintained during the 28-week extension period. In addition, henagliflozin showed a good tolerability profile, with high medication compliance over the 52-week treatment.

In our study, the placebo-adjusted LS mean change in HbA1c was -0.91% and -0.94% for henagliflozin 5 and 10 mg, respectively, after 24 weeks of treatment. This magnitude of reduction was similar to other SGLT2 inhibitors approved as monotherapy in predominantly Asian or non-Asian patients with T2D.^{14–18} In addition, the glycaemic efficacy of henagliflozin was evident across all subgroups and was sustained over the 52-week study period, highlighting the broad spectrum of patients who could derive durable clinical benefit. The greater reduction in HbA1c with both doses of henagliflozin over placebo was accompanied by a higher proportion of patients achieving the HbA1c goal of less than 7%, as well as improvements in FPG, 2-h PPG and PPG AUC_{0–2h}, consistent with observations that SGLT2 inhibitors impact both fasting and postprandial glucose levels.^{16,17} Notably, both doses of henagliflozin produced a similar reduction in HbA1c and plasma glucose in the overall population. Nevertheless, our subgroup analysis showed a trend of greater reduction in HbA1c for henagliflozin 10 mg over 5 mg in patients with a baseline HbA1c of

more than 8.5%. With their unique insulin-independent mechanism of action, SGLT2 inhibitors have been widely shown to be effective both as monotherapy or as combined therapy with other glucose-lowering drugs for treatment of T2D.¹⁹ The role of henagliflozin as an add-on therapy to metformin will be further elucidated in a separate phase 3 trial (NCT04390295).

Obesity and high blood pressure are well-established CV risk factors and an important part of multifactorial T2D management.²⁰ Henagliflozin 5 and 10 mg significantly reduced body weight by 1.3 and 1.5 kg, respectively, from baseline at Week 24 compared with placebo. This magnitude of reduction was comparable with the weight loss observed for dapagliflozin in a predominantly Chinese T2D population.¹⁴ Importantly, weight loss with SGLT2 inhibitors was mainly attributed to the reduction in fat mass/visceral fat mass,²¹ which could be especially relevant in Asian patients who are characterized by a higher proportion of visceral fat mass.²² Further, 24-week treatment with henagliflozin resulted in a clinically meaningful reduction in both SBP (5/10 mg: $-5.1/-4.4$ mmHg) and DBP (5/10 mg: $-3.0/-1.7$ mmHg) versus placebo, without causing hypotension events. These reductions compared favourably with the reductions of 1.2–2.3 mmHg in SBP and of 1.3–1.6 mmHg in DBP reported for dapagliflozin in predominantly Chinese patients with T2D.¹⁴ The antihypertensive mechanisms of SGLT2 inhibitors are not entirely understood, but could involve increased urinary sodium excretion and improved vascular endothelial function.²³ The underlying mechanism for the antihypertensive effects observed for henagliflozin requires further investigation. Recently, the pivotal DAPA-HF and EMPEROR-Reduced trials showed that SGLT2 inhibition could reduce the risk of CV death or hospitalization for heart failure in patients with heart failure.^{10,24} The current study has confirmed that the cardiometabolic improvement in weight loss and blood pressure with henagliflozin was sustainable, and the longer term benefits in possible reduction of CV morbidity/mortality should be explored in the future, especially in patients with high CV disease risk.

Both doses of henagliflozin were well tolerated during the 52-week study with no increased risk of AEs noted with prolonged drug exposure. The most frequent AEs for henagliflozin were ketosis-related events, which could be expected from the metabolic shift towards lipids and ketone bodies as energy fuel resulting from increased urine glucose excretion by the drug.²⁵ Importantly, most cases of ketosis were mild, with no case of DKA observed in our study, consistent with the low occurrence rate of DKA documented for other SGLT2 inhibitors.¹⁹ The incidence of urinary tract infections was generally low and similar across the henagliflozin and placebo groups, consistent with a meta-analysis of 86 randomized trials reporting that the risk of urinary tract infections was not increased with SGLT2 inhibitors compared with placebo or active control.²⁶ Notably, the incidence of genital infections, a common AE reported for SGLT2 inhibitors, was also low ($<2\%$) across all treatment groups. This was probably attributable to the personal hygiene advice we provided to participants before initiation of study treatment, suggesting that hygiene education may help to reduce the occurrence of genital

infections in patients treated with an SGLT2 inhibitor and without a prior history of recurrent genital infections. Similar to reports for other SGLT2 inhibitors,⁵ the incidence of hypoglycaemia was low with henagliflozin monotherapy (1.3%–2.6%) and no major episodes of hypoglycaemia were observed, showing the advantage of an insulin-independent mechanism of action.

The current study has some limitations. First, because T2D is a chronic and progressive disease, a longer term study beyond 52 weeks may be needed to fully assess the efficacy (e.g. the cardio-renal effects), durability and safety of henagliflozin. Second, the comparative efficacy (glycaemic and non-glycaemic) and safety of henagliflozin versus other antidiabetic agents such as dipeptidyl peptidase-4 inhibitors should be explored in future active-controlled studies.

In summary, henagliflozin monotherapy (5 and 10 mg) provided effective glycaemic control, reduced body weight and blood pressure, and was generally well tolerated in Chinese patients with T2D inadequately controlled on diet and exercise.

ACKNOWLEDGEMENTS

We thank the patients and their families for their participation in this study, as well as the study teams at all trial centres. We would also like to acknowledge Xiuzhi Wu, PhD (a medical writer at Hengrui) for medical writing support according to Good Publication Practice Guidelines. The study was sponsored by Jiangsu Hengrui Medicine Co., Ltd.

CONFLICT OF INTEREST

HL, TZ and JZ were all employees of Hengrui Medicine at the time this study was conducted. All the other authors declare that there are no conflicts of interests.

AUTHOR CONTRIBUTIONS

Design: JL, HL and JZ; conduct/data collection: JL, LF, YL, JG, LQ, PL, HZ, ZS, YL, LZ, YS, DC, GQ, WL, YG and YZ; statistical analysis: TZ; interpretation of data: all authors; drafting of the manuscript: JL; critical revision of the manuscript for important intellectual content: all authors; approval of the final manuscript: all authors.

DATA AVAILABILITY STATEMENT

All data analyzed during this study are included in this published article or as supplementary information files.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

How to cite this article: Lu J, Fu L, Li Y, et al. Henagliflozin monotherapy in patients with type 2 diabetes inadequately controlled on diet and exercise: A randomized, double-blind, placebo-controlled, phase 3 trial. *Diabetes Obes Metab*. 2021; 1-10. <https://doi.org/10.1111/dom.14314>